Fine-tuning LLMs for Text Summarization

BART-base vs GPT-2 on CNN/DailyMail | Santiago Rodriguez Ramirez

Two pre-trained models below 1B parameters were fine-tuned end-to-end on the CNN/DailyMail v3.0.0 abstractive summarization task: an encoder-decoder (BART-base, 139.4M params) and a decoder-only (GPT-2, 124.4M params). Both share the same recipe (AdamW at 3e-5, 10% linear warmup, weight decay 0.01, FP16, batch size 8 with 4 grad-accumulation steps for an effective batch of 32, 3 epochs, 50,000-article train subset, 5,000-article validation subset, 4-beam search with no_repeat_ngram_size = 3, max 128 generated tokens). All training ran on two Tesla T4 GPUs on Kaggle. Final evaluation uses stemmed ROUGE-1/2/L on a 3,000-article slice of the official test split.

1. Final Configurations and Headline Result

BART-base (encoder-decoder). Input: "summarize: " + article, truncated to 512 sub-words. Target: highlights, truncated to 128 sub-words. Cross-entropy on the decoder labels (DataCollatorForSeq2Seq pads labels with -100). Chosen because BART's denoising pre-training (text infilling, sentence permutation) is structurally close to summarization, and its bidirectional encoder plus cross-attending decoder are the canonical architecture for conditional generation.

GPT-2 (decoder-only). Article and summary concatenated under the "instruction" prompt template ("Summarize the following news article.\n\nArticle: {article}\n\nSummary: {summary}<|endoftext|>"), truncated to 1024 tokens. Loss is masked (-100) on the article and prompt tokens so cross-entropy applies only to the summary span. Padding side = left, pad_token = eos, custom collator. Chosen as the canonical small decoder-only baseline at the same scale as BART-base, which isolates the architectural comparison from a model-size confound.

Model	Params	ROUGE-1	ROUGE-2	ROUGE-L
BART-base (encoder-decoder)	139.4M	41.28	18.69	27.98
GPT-2 (decoder-only)	124.4M	32.37	8.50	19.31
Delta (BART vs GPT-2)	n/a	+8.91	+10.19	+8.67

2. What Was Explored and Why

Three design dimensions were ablated. They cover the data, the optimiser, and the prompt: the three places this kind of fine-tuning project most often goes wrong.

- Training-set size (5k, 15k, 30k samples). Separates sample-efficiency from peak performance and exposes saturation.
- Learning rate (1e-5, 3e-5, 5e-5 at 15k samples). The optimiser's most influential hyperparameter; also tests whether a single recipe transfers between architectures.
- Prompt format for GPT-2 (simple, instruction, structured). Decoder-only only; for that family the prompt is part of the architecture.

Other levers were considered but rejected: model size (would confound architecture against capacity), context length (the 512/128 split already covers the median article and 95th-percentile summary), and sampling strategy (4-beam search dominates nucleus and temperature for ROUGE on news summaries).

3. Ablation Results and Analysis

Ablation runs use a fresh checkpoint per configuration, 2 epochs, a 500-article validation slice for loss-based evaluation, and a 1,000-article test slice (a strict subset of the headline 3k slice; absolute ROUGE values are directly comparable, with slightly higher variance).

3.1 Training-set size

Model	Train size	ROUGE-1	ROUGE-2	ROUGE-L
BART-base	5,000	39.98	17.93	27.18
BART-base	15,000	40.80	18.42	27.67
BART-base	30,000	40.55	18.29	27.43
GPT-2	5,000	25.65	5.94	16.73

Model	Train size	ROUGE-1	ROUGE-2	ROUGE-L
GPT-2	15,000	31.80	8.08	19.77
GPT-2	30,000	32.44	8.42	19.39

BART saturates fast. +0.82 ROUGE-1 from 5k to 15k, then a 0.25-point dip at 30k, consistent with mild overfitting against the 500-sample validation set. **GPT-2 is data-hungry.** +6.15 ROUGE-1 from 5k to 15k and +0.64 more at 30k. The crucial point is that GPT-2 at 30k samples (32.44 R1) is still about 8 points below BART at 5k samples (39.98 R1). At this scale, the encoder-decoder inductive bias (bidirectional encoding of the article plus decoder cross-attention back to it) delivers far more lift than any reasonable extra data we can give a same-size GPT-2.

3.2 Learning rate

Model	Learning rate	ROUGE-1	ROUGE-2	ROUGE-L
BART-base	1e-5	40.64	18.29	27.54
BART-base	3e-5	40.80	18.42	27.67
BART-base	5e-5	40.73	18.37	27.42
GPT-2	1e-5	26.36	6.28	17.00
GPT-2	3e-5	31.80	8.08	19.77
GPT-2	5e-5	(not completed)	n/a	n/a

BART is essentially LR-insensitive in this band (40.64, 40.80, 40.73; spread under 0.16 R1). **GPT-2 is highly LR-sensitive.** 1e-5 underfits badly within 2 epochs (26.36 R1, similar to the 5k-data score), while 3e-5 reaches 31.80. The 5.4-point gap from the optimal LR is the largest single hyperparameter effect in the whole study. The GPT-2 5e-5 cell ran but the Kaggle 9-hour wallclock killed the session before its result line printed; the same compute exhaustion is also why the prompt-format ablation below could not be executed. The visualisation notebook explicitly logs this.

3.3 Prompt format (GPT-2 only): not executed

Three templates were defined: simple ("Article: ... Summary: ..."), instruction (an explicit imperative), and structured ("### Article ... ### Summary"). The runs were implemented but did not execute due to the wallclock limit. Expected outcome: instruction wins because the imperative gives GPT-2 an unambiguous task signal; simple is a close runner-up because it leaves more of the 1024-token context window for the article (median article around 750 to 900 tokens); structured is the riskiest because markdown headers cost tokens without adding a clear instruction signal. Across all three, the ROUGE delta is likely small (about 1 to 2 R1) since loss masking and the prediction span are identical: the prompt only changes what the model conditions on.

4. Takeaways

Architecture is the dominant lever (about +9 R1 BART vs GPT-2). It outweighs every other dimension we tested combined.

- **Recipe portability.** AdamW at 3e-5, 10% warmup, FP16, effective batch 32 is safe for both models at this scale. BART tolerates a 5x LR range while GPT-2 silently loses about 5 R1 at 1e-5, so an LR sweep is required when scaling GPT-2-style models.
- **Compute-efficiency winner.** BART-base trained on 5k samples (39.98 R1) already beats GPT-2 trained on 30k samples (32.44 R1). Architecture choice is worth more than 6x the data.
- **Caveats.** Single seed (42); the 0.25 R1 BART dip at 30k and the 0.07 R1 BART LR spread are within typical seed-to-seed noise; ablation runs are 2 epochs vs 3 in the headline (slight under-estimation); GPT-2 LR=5e-5 and the prompt-format ablation did not finish (Kaggle 9-hour wallclock). All experimental scaffolding is in place to re-run on a longer session.